

TECHNICAL POSITION PAPER

UDP Inter-AS Without Session State:

A Structural Attack Vector on the Internet and the Case for Its Non-Routability

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Executive Summary

This document argues that the User Datagram Protocol (UDP), in its form without verifiable session state, should not be routable between Autonomous Systems (ASes) on the public internet. The central argument is not technical-protocolar but structural: UDP without a prior handshake constitutes a systemic damage asymmetry where the cost of the attack is minimal for the origin and catastrophic for the destination, violating the reciprocity principle that underpins network neutrality.

The document presents two complementary lines of argument: (A) the maximalist position of inter-AS non-routability of raw UDP, and (B) the compromise position that requires verifiable session state as a condition of routability, acknowledging that protocols such as QUIC, WireGuard, and DTLS already meet this requirement. The role of IXPs as the enforcement point is argued to be central to any viable solution.

1. Introduction

For 25 years, the network operator community has watched UDP amplification traffic destroy infrastructure, consume transit bandwidth, and force smaller operators to pay millions for mitigation services or simply go down. The standard response has been BCP38, published in May 2000 — a set of voluntary recommendations that, as of this document, still lack universal adoption.

The question this document asks is not 'how do we mitigate malicious UDP' but 'why does the structural condition that makes it possible exist, and what fundamental change would eliminate it'.

The author operates an Internet Exchange Point with infrastructure in Chile, Mexico, Argentina, and Peru, and writes from the perspective of someone managing the exact point where this decision would need to be made.

2. Technical Background

2.1 UDP vs TCP: The structural difference

TCP (Transmission Control Protocol, RFC 793) requires a three-way handshake before transmitting data. This mechanism has a fundamental security consequence: the origin must respond to the destination's SYN-ACK. If the source IP is spoofed, the SYN-ACK reaches the fake IP and the connection never completes. The attacker cannot generate valid traffic without controlling the real origin.

UDP (User Datagram Protocol, RFC 768) is a connectionless protocol. It requires no handshake, maintains no state, and does not verify that the receiver is expecting the traffic. A UDP packet can arrive from any IP — real or spoofed — and the receiver must process or discard it, but it already received it, already consumed bandwidth, already used resources.

Feature	TCP	UDP
Handshake	3-way (SYN/SYN-ACK/ACK)	None
Origin verification	Implicit via handshake	Nonexistent
Session state	Mandatory	Nonexistent
Amplification possible	No (1:1 maximum)	Yes (up to 50,000:1)
Effective origin spoofing	No	Yes
Congestion control	Built-in	Nonexistent

2.2 The UDP amplification mechanism

UDP amplification attacks exploit servers that respond to queries with responses much larger than the original query. The attacker sends a small query with a spoofed source IP (the victim), and the server responds with a large payload directly to the victim.

Documented amplification factors by protocol:

- DNS: up to 28:1 (ANY query)
- NTP (monlist): up to 556:1
- Memcached: up to 51,000:1 (2018 attack on GitHub: 1.35 Tbps)
- SSDP: up to 30:1
- CharGen: up to 358:1

The damage formula is simple: if an attacker has 1 Gbps and uses a vulnerable NTP server, they can generate 556 Gbps at the victim. No medium or small operator has the capacity to absorb that. The only defenses are paying a scrubbing provider, configuring a blackhole (which kills the service), or waiting for the attack to stop.

2.3 BCP38: 25 years of failed voluntarism

BCP38 (RFC 2827, published May 2000) defines ingress filtering to block packets with spoofed source IPs. If all operators implemented BCP38, UDP amplification would be impossible because the spoofed packet would never leave the origin network.

The problem is that BCP38 is voluntary and adoption remains partial. According to CAIDA's Spoofer project data, approximately one quarter of observed networks allow full or partial spoofing — projected across the internet, that equals more than 360 million addresses from which spoofing is possible. This is 25 years after the RFC was published.

The reason for non-adoption has a name: tragedy of the commons. The operator who implements BCP38 incurs an operational cost with no direct benefit. The benefit goes to third parties. In a maximum-margin market, nobody implements what generates no revenue.

2.4 State of the Art at IETF: SAVNET and BAR-SAV

Before presenting this document's proposal, it is necessary to acknowledge that the IETF has not been completely inactive on the inter-domain spoofing problem. There is ongoing work that approaches the problem — but from a different layer.

SAVNET (Source Address Validation Network) is an active IETF working group with inter-domain architecture drafts for AS-to-AS source IP validation. The architecture defines SAV-specific communication mechanisms between ASes so that each learns the legitimate prefixes and valid ingress addresses, automatically distributing that information when routes change.

BAR-SAV (Source Address Validation Using BGP UPDATEs, ASPA, and ROA) is an active draft from NIST and Akamai, version 08 as of October 2025, proposing SAV filters using BGP UPDATE messages, ASPA and ROA to minimize false positives in inter-domain origin validation.

Both are serious and necessary work. But they address the problem from L3: who can announce which prefix, and from which AS can a packet with a given source IP come. Neither touches session state at the transport protocol level.

The difference is fundamental:

Mechanism	Layer	What it validates	Attack vector addressed	IETF status
BAR-SAV / SAVNET	L3 (source IP)	Prefix vs announcing AS	Origin spoofing	Active draft 2025
This proposal (Line B)	L4 (UDP session)	Prior verifiable handshake	UDP amplification without session	Does not exist

They are complementary, not equivalent. BAR-SAV can validate that the packet comes from who it claims to be. This proposal goes one step further: it requires that there be a valid reason for that UDP packet to be traveling inter-AS in the first place. A UDP amplification packet can have a valid source IP and still be an attack. BAR-SAV does not stop it. Line B does.

The fact that the IETF is working on SAVNET and BAR-SAV confirms that the community has already recognized that voluntary BCP38 is insufficient. This document proposes the next logical step: if we are going to require that the origin be valid, we should also require session context. One condition without the other is a half-solution.

3. The Central Argument: Network Neutrality vs Transport Anarchy

Network neutrality is a principle that protects content and applications: an ISP cannot discriminate Netflix traffic over YouTube, or block VoIP to protect its own telephony service. This principle is correct and necessary.

But network neutrality was never designed to protect attack vectors. The freedom to carry traffic on the internet is not and should not be equivalent to the freedom to consume a third party's bandwidth at no cost and with no reciprocity.

UDP without session state on the public internet is not 'neutral traffic' — it is a structural damage asymmetry. The attacker spends a fraction of the resource they destroy at the destination. There is no reciprocity, no proportional cost, no accountability mechanism.

This asymmetry is not accidental. It is a direct consequence of UDP's design in 1980, when hardware was limited and the internet was an academic network of mutual trust. The world of 2026 is different: the internet is global critical infrastructure and stateless UDP is the dominant volumetric attack vector.

3.1 The critical distinction: firewall vs non-routability

The traditional solution to malicious UDP traffic is destination filtering: a firewall, an ACL, a scrubbing center. This approach has a fundamental flaw that is rarely discussed: the damage has already occurred.

When a firewall drops a malicious UDP packet, that packet has already:

- Consumed transit bandwidth on every AS that forwarded it
- Occupied capacity at the IXP where it was routed
- Reached the destination AS border
- Been processed by hardware (even if just to discard it)

```
FIREWALL:      [attacker] --> [internet] --> [dest border] --> DROP
                ^-- transit BW already consumed
```

```
NON-ROUTABLE: [attacker] --> [first AS without UDP routability] --> DROP
                ^-- traffic dies here, never travels
```

The difference is that in the non-routability model, transit bandwidth is not consumed, the IXP does not process it, and the destination never sees it. The damage does not exist. This is the central argument of this document.

4. Line A: UDP Should Not Be Routable Inter-AS (Maximalist Position)

The maximalist position holds that UDP, as a stateless protocol, should not appear in routing tables between Autonomous Systems. Its use would be restricted to private networks (LAN, corporate WAN, labs) where the operator controls all endpoints and can guarantee the context of use.

4.1 Technical basis

If UDP is not routable between ASes, the following attack vectors disappear by design, not by configuration:

- DNS/NTP/Memcached/SSDP amplification: impossible without inter-AS UDP routability
- Origin spoofing with reflection: the spoofed packet never leaves the origin AS
- Direct UDP flood from inter-AS botnet: not routable

4.2 What survives in LAN/private

Restricting UDP to private networks does not eliminate legitimate use cases that do not require the public internet:

- Internal DNS (resolution within the AS or corporate network)
- DHCP (LAN protocol by design)
- Internal NTP
- RTP/VoIP within private networks or with direct peering
- LAN gaming
- Internal multicast

4.3 The big problem: QUIC and HTTP/3

The most serious objection to Line A is QUIC. HTTP/3 runs over QUIC, which runs over UDP. Adoption numbers are significant: by 2024, approximately 20% of global web requests used HTTP/3, and platforms like Meta report that more than 75% of their internet traffic already uses QUIC/HTTP/3.

If UDP is not routable inter-AS, HTTP/3 falls. That is not trivial: it means reverting to the HTTP/2+TCP ecosystem at a moment when the industry has invested enormously in the transition to QUIC for its performance and security advantages.

This is the argument that makes Line A, though technically correct in its diagnosis, practically unviable without a planned transition. That is why Line B exists.

5. Line B: UDP With Verifiable Session State as a Routability Condition

The compromise position recognizes that the problem is not UDP as a protocol but UDP without prior session context. If someone is forwarding legitimate UDP inter-AS, it is because there was a prior handshake or negotiation that establishes the context of the communication. That handshake is verifiable.

The reformulation of the principle is:

"UDP inter-AS without prior verifiable session state should not be routable. UDP with documented handshake is legitimate because the origin has already been validated."

5.1 UDP protocols that meet the criterion

Protocol	Handshake	Meets criterion	Notes
QUIC/HTTP3	TLS 1-RTT / 0-RTT	Yes	Cryptographic handshake
WireGuard	Noise IK handshake	Yes	Mutual auth before data
DTLS	Similar to TLS	Yes	IETF standard RFC 9147
DNS (query)	None	No	Amplification vector
NTP (monlist)	None	No	Amplification vector
Raw UDP	None	No	Drop at inter-AS border
RTP/SRTP	Negotiated via SIP/SDP	Conditional	Requires prior signaling

5.2 How the model works

The enforcement model at the AS border would be as follows:

- Inter-AS UDP is dropped by default at border routers
- Inter-AS UDP is permitted if it belongs to a flow with a prior verified handshake (QUIC Connection ID, WireGuard session, DTLS session)
- Border devices maintain a table of authorized UDP session states, similar to how a stateful firewall handles TCP
- UDP without session context (raw DNS, raw NTP, arbitrary UDP) does not cross AS boundaries

5.3 QUIC as a bridge between Line A and Line B

QUIC is precisely the answer the industry is already building, albeit without the regulatory framework this document proposes. QUIC solves the problems of raw UDP:

- Integrated TLS 1.3 cryptographic handshake: the origin must respond
- Connection ID: allows identifying sessions and validating that traffic belongs to an established session
- Integrated congestion control: eliminates the network flooding problem
- Mandatory encryption: there is no QUIC without TLS

If the operator community adopts Line B, QUIC and HTTP/3 survive unchanged. WireGuard survives. DTLS survives. What falls is exactly what should fall: raw UDP without context, which is the amplification vector.

6. The Technical Solution: Implementation at the AS Border

6.1 Enforcement point

The correct enforcement point is the AS border, not the destination. Specifically:

- Border routers of the AS facing the IXP or transit
- IXP route servers as a complementary policy point
- Not in the AS core or internal routers

The IXP is the architecturally correct point because:

- It is where ASes interconnect: blocking here means the traffic never travels
- The IXP already enforces routing policies (route servers, prefix filters): adding transport policy is a natural extension
- An IXP that implements this simultaneously protects all its members
- The IXP operator has direct incentive: less malicious traffic means lower operational cost for everyone

6.2 Technical mechanism

Line B implementation (UDP with verifiable session):

Component	Description
Stateful UDP tracking	Border router maintains table of active UDP sessions identified by QUIC Connection ID, WireGuard session key hash, or DTLS session ID
Default deny inter-AS UDP	Border ACL: deny udp any any by default on interfaces toward IXP and transit
Permit by known protocol	permit udp any any match quic-handshake / wireguard-handshake / dtls-hello
Session timeout	Authorized UDP sessions expire with no traffic after N seconds (configurable, default 30s)
Logging and telemetry	All dropped UDP is logged with src/dst IP, port, origin AS for analysis

6.3 Special cases requiring explicit treatment

The following UDP protocols require specific treatment during the transition:

- DNS over UDP/53: migrate to DNS over TCP (already supported), DNS over TLS (DoT, port 853), or DNS over HTTPS (DoH, port 443). The migration is technically mature and most modern resolvers support it.
- NTP: migrate to NTPsec with authentication, or encapsulate in TCP/TLS. NTPsec eliminates monlist by design.
- RADIUS/DIAMETER: encapsulate in DTLS or migrate to RadSec (RADIUS over TLS).
- SNMP: v3 over UDP with authentication and encryption meets the verifiable session criterion; v1/v2c do not.
- Online gaming and real-time streaming: QUIC or DTLS depending on the case.

6.4 Existing tools and why they are not enough

Before presenting Line B as a new proposal, it is necessary to acknowledge the tools the industry already uses to address this problem — and be honest about their limitations.

uRPF (Unicast Reverse Path Forwarding)

uRPF validates that the source IP of a packet is reachable via the interface it arrived on. Two modes exist: strict (packet must return via the same interface) and loose (only verifies that the source IP exists in the routing table). Strict is effective but breaks networks with asymmetric routing, which is the norm on the internet. Loose is useful but does not block spoofing of valid IPs that simply do not belong to the sending AS.

uRPF is the technical implementation of BCP38. It has been best practice for 25 years and a quarter of networks still have not implemented it. This is not a tool problem — it is an incentive problem. BAR-SAV is the more precise successor using RPKI/ASPA. The conclusion does not change: all of these attack L3, none touches UDP session state at L4.

Static ACLs by port

Blocking UDP ports known as amplification vectors: UDP/11211 (memcached), UDP/389 (CLDAP), UDP/1900 (SSDP), UDP/19 (chargen), UDP/123 (NTP monlist). This is practiced today at many ISPs and IXPs and is useful as a transition measure.

The problem is that it is whack-a-mole. Every time a new amplification protocol appears, a new ACL must be added. The attacker only needs to find a protocol not on the list. This is exactly the counterargument that reinforces Line B: instead of maintaining a blacklist of bad ports (reactive), Line B proposes a whitelist of valid sessions (proactive).

BGP Flowspec

Flowspec (RFC 5575, updated RFC 8955) is the most interesting of the three tools. It allows distributing filtering rules via BGP: an operator can announce to its peers a rule like 'drop all stateless UDP from this prefix' and the peers' border routers apply it automatically.

Flowspec is the enforcement mechanism closest to what this document proposes. In fact, the IXP2 model could use Flowspec as the technical vehicle to distribute the UDP non-routability policy among certified peers, without needing a Route Server.

But it has critical limitations in its current use:

- Reactive: someone must detect the attack first and then announce the Flowspec. Not preventive by design.
- Partial adoption: not all border routers support it or have it enabled in production.
- Limited scope: generally used for mitigation of ongoing attacks, not as a permanent session policy.
- Short lifetime: Flowspec rules typically have a timeout and require manual or automated renewal.

This document proposes using Flowspec as a permanent policy vehicle, not reactive mitigation. An IXP that distributes via Flowspec the rule 'deny inter-AS UDP without verifiable session' to all certified members is implementing Line B with tools that already exist today on most modern border equipment.

Tool	Layer	Type	Main limitation	Role in Line B
uRPF strict/loose	L3	Reactive / preventive	Breaks asymmetric routing; does not touch UDP session	Complementary, not sufficient
Static ACLs by port	L4	Reactive	Blacklist: attacker finds new port	Transition measure, not solution

Tool	Layer	Type	Main limitation	Role in Line B
BGP Flowspec	L3/L4	Reactive	Used to fight fires, not prevent them	Ideal vehicle to distribute Line B policy
Line B (this proposal)	L4	Preventive	Requires inter-operator agreement and stateful hardware	The missing layer

7. Costs, Benefits, and Counterarguments

7.1 Implementation costs

Component	Estimated cost	Notes
Stateful border hardware	High for legacy equipment	Modern equipment (Cisco, Juniper, Nokia) supports stateful UDP in hardware
DNS migration to DoT/DoH	Medium	Requires resolver and authoritative reconfiguration
Operator training	Low-Medium	Configuration paradigm shift
Legacy application updates	High in specific cases	Applications using raw UDP without handshake require redesign
Inter-AS coordination	High	Requires inter-operator policy agreement

7.2 Benefits

Benefit	Estimated impact
Elimination of UDP amplification	Eliminates the dominant volumetric DDoS vector (60-80% of attacks by volume)
Reduced mitigation cost	Medium and small operators stop paying for massive scrubbing
Traffic traceability	All inter-AS UDP has identifiable session context
Reduced junk transit traffic	Transit bandwidth not consumed by malicious traffic
Incentive for QUIC adoption	Accelerates transition to HTTP/3 and modern protocols
Protection for small operators	Most affected by volumetric DDoS are those with fewest resources to mitigate it

7.3 Counterarguments

7.3.1 'QUIC solves the problem without changing routability'

Counterpoint: QUIC solves the problem for new applications but does not eliminate the existing universe of raw UDP. While DNS, NTP, and thousands of legacy applications continue using UDP without a handshake, the vector exists. Line B proposes exactly this: QUIC is the path, but it must be a condition, not a recommendation.

7.3.2 'IXPs cannot impose Layer 4 policies'

Counterpoint: IXPs already impose Layer 3 policies (prefix filters, RPKI). Extending to Layer 4 is a policy decision, not a technical limitation. Modern route servers support Layer 4 filters. Furthermore, the policy can be implemented on each member's border routers without the IXP needing to inspect traffic.

7.3.3 'Violates network neutrality'

Counterpoint: Network neutrality protects content and applications, not attack vectors. A protocol that by design cannot be distinguished from an attack does not have the same protection as an application that uses that protocol in a verifiable way. Line B does not discriminate by content but by the presence or absence of session state — the same thing TCP does by design.

7.3.4 'Migration cost is prohibitive'

Counterpoint: The cost of not migrating is higher. UDP amplification attacks cost the industry billions of dollars annually in mitigation, idle capacity, and service damage. The question is not whether migration cost is high but whether it is greater than the accumulated cost of inaction.

7.3.5 'IPv6 complicates implementation'

Counterpoint: IPv6 actually strengthens the argument. Without NAT, every device has a public IP, which increases the UDP amplification attack surface. Line B implementation in IPv6 is technically identical to IPv4.

8. The Future of the IXP and the Death of the Route Server

There is an elephant in the room this document cannot ignore: Route Servers — the centralized policy mechanism in many IXPs — are in decline. The clear trend in the operator community is toward direct AS-to-AS peering, eliminating dependence on the RS as a centralized control point.

There are operational reasons for this trend. The RS is a potential single point of failure. It introduces additional latency in route negotiation. Large operators prefer direct control over their BGP sessions. And as the BGP automation ecosystem matures, the utility of the RS as an operational simplifier diminishes.

This requires rethinking the enforcement model in this document's proposal. And the conclusion is that the death of the RS actually strengthens the argument, not weakens it, because it forces a more robust and decentralized solution.

8.1 The IXP2 model without Route Server

If peering is direct AS-to-AS, enforcement cannot be centralized. The mechanism shifts from technical control to certification and governance.

The IXP does not filter. The IXP certifies.

The IXP maintains a public certification registry, extensible over PeeringDB or equivalent. Members that implement UDP stateful on their border declare, demonstrate, and become certified. When AS-A (certified) negotiates direct peering with AS-B, it queries the IXP registry. If AS-B is not certified, AS-A accepts the peering but applies local policy: drop incoming sessionless UDP from that peer. Autonomous decision by each AS, informed by the IXP registry.

```
IXP
├── CERTIFICATION REGISTRY (public, like PeeringDB)
│   ├── member declares and implements UDP stateful at border
│   └── IXP verifies and publishes certified list --> Plane A
└── PLANE B: Standard
    ├── no certification, connects as usual
    ├── certified peers apply local policy
    └── on sessionless UDP from non-certified peers
```

8.2 Why this model is more robust

A RS-based model has a concentration problem: if the RS fails or does not filter, the policies fail with it. A model based on IXP certification + local policy at each AS is distributed by design. Each AS applies the policy at its own border, informed by the centralized registry. The registry can have problems and ASes continue filtering because the policy lives locally.

It is the same principle as RPKI: the ROA repository can fail, but routers that have already validated the ROAs continue filtering. Validation is distributed even though the repository is centralized.

8.3 Gradual migration without exclusion

A small ISP that cannot implement UDP stateful at its border is not excluded from the IXP. It connects as usual, in Plane B. The difference is that certified peers can apply a more restrictive policy on incoming UDP. Over time, when the ISP updates its hardware or configuration, it requests certification and moves to Plane A. The migration is organic, not imposed.

This is exactly the model RPKI followed in IXPs. First voluntary, then incentivized because certified peers are more trustworthy, eventually de facto standard. There was no exclusion. There was differentiation.

9. The Role of IXPs as Enforcement Points

Internet Exchange Points are architecturally the correct point to implement this policy for several reasons that go beyond the technical.

9.1 Position in the topology

The IXP is the point where multiple ASes interconnect multilaterally. A UDP packet crossing the IXP between two member ASes is exactly the traffic this document seeks to control. The IXP

has visibility and policy capability over that traffic without requiring bilateral coordination between every pair of ASes.

9.2 Existing precedent

IXPs already enforce routing security policies:

- Prefix filters on route servers
- RPKI ROV (Route Origin Validation) at some IXPs
- Policies against announcing invalid routes
- Escalation procedures for anomalous member behavior

Adding a UDP transport policy is not a conceptual leap — it is an extension of the existing framework.

9.3 IXP incentive

Unlike an individual ISP where BCP38 is a cost with no direct benefit, the IXP has a clear incentive: less malicious traffic means lower load on the IXP fabric, lower capacity cost, and a more attractive service for members. The IXP that can say 'sessionless UDP does not cross our fabric' offers a real value proposition.

9.4 Coordination with MANRS

MANRS (Mutually Agreed Norms for Routing Security) already establishes a framework of voluntary norms for routing security at IXPs. This document's proposal is complementary: while MANRS addresses L3 security (routing, prefixes, RPKI), a natural extension would be an L4 norm requiring that members not forward inter-AS UDP without verifiable session state.

By 2024, MANRS had 1,012 participants in the NetOps category and 31 participating IXPs. A formal proposal in this framework would have real reach. And given the trend toward RS elimination, MANRS will at some point need to address how its norms are implemented in a world of direct peering. This document is a concrete proposal for that question.

10. Concrete Proposal

10.1 Immediate action: IXPs as early adopters

MANRS member IXPs should adopt as voluntary policy the following rule on their route servers and recommend its implementation by members on border routers:

- Drop inter-AS UDP without a verifiable QUIC Connection ID, WireGuard session, or DTLS session ID
- Allow DNS over UDP/53 with aggressive rate limiting as a transition measure (max 1000 pps per member)
- Mandatory logging of dropped UDP for visibility and analysis

10.2 Medium-term action: IETF proposal

Present an Internet Draft to the IETF Transport Area (TSVWG) that defines:

- A formal classification of inter-AS UDP into: UDP with verifiable session state (permitted) and UDP without session state (not recommended for inter-AS routing)

- Technical criteria for a UDP protocol to qualify as 'verifiably stateful'
- Implementation recommendations for border routers

10.3 Long-term action: regulatory policy

In the context of LACNIC/LACNOG and regional telecommunications regulators, propose that the interconnection condition at IXPs include the commitment not to forward inter-AS UDP without verifiable session state, as an equivalent condition to BCP38 adoption.

This requires no new legislation: it can be articulated as a peering condition in existing interconnection agreements, in the same way some IXPs already condition peering on RPKI adoption.

11. Conclusions

Stateless UDP on the public internet is a design flaw whose consequences were unpredictable in 1980 but are perfectly clear in 2026. The protocol was not designed for the internet that exists today, and the voluntary solution (BCP38) has demonstrated over 25 years that voluntarism does not work.

This document does not propose killing UDP. It proposes that UDP, to be routable between Autonomous Systems, must meet the same basic condition that TCP meets by design: that the origin is verifiable through an established session state.

Line A (total non-routability) exists as a maximalist argument to open the debate. Line B (UDP with verifiable session) is the practically viable proposal, and QUIC is the demonstration that the industry is already heading in that direction — the only thing missing is the framework that makes it a condition rather than a recommendation.

The IXP2 model resolves the objection that the RS is going to die. It does not need a RS. It needs a certification registry, local policy at each AS, and a community willing to differentiate between peers that do the work and those that do not. It is the same model that worked with RPKI.

IXPs are the correct enforcement point. They have the architectural position, the economic incentive, and the policy precedent to lead this change.

The internet needs someone to ask the uncomfortable question. This document is that question.

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Appendix A: Action Plan

IXP implementation and member adoption strategy

This appendix is operational. It is not theory — it is what an IXP needs to do to implement this document's proposal, and how to get its members to adopt it. It is separated from the body of the paper because the tone is different: it is a work plan, not an academic argument.

A.1 Phase 1: The IXP gets its own house in order first

You cannot ask members to implement what the IXP has not done first. The credibility of the proposal depends on PIT Chile (or the leading IXP) having its own house in order before speaking to anyone.

A.1.1 Internal audit

Before doing anything external, the IXP must know exactly what it has:

- Border equipment inventory: which models, which software versions, whether they support stateful UDP tracking and BGP Flowspec in hardware.
- Current uRPF status: how many member ports have uRPF enabled, in which mode (strict/loose), how many do not.
- Flowspec status: whether the IXP already distributes Flowspec rules to members, with what coverage, whether any members do not support it.
- Current UDP traffic: baseline of inter-AS UDP volume on the fabric, port distribution, top sources. This is the post-implementation comparison point.

A.1.2 Implementation on IXP border routers

The IXP implements the Line B policy on its own border routers first as internal policy:

- Border ACL: deny inter-AS UDP by default on member-facing interfaces, with explicit permit for verifiable QUIC, WireGuard, DTLS sessions.
- Aggressive rate limiting on DNS/UDP/53: max 1000 pps per member as a transition measure while they migrate to DoT/DoH.
- Complete logging of dropped UDP: src/dst IP, port, origin AS, volume. This telemetry is the most powerful argument for convincing members — numbers speak.
- Internal Flowspec: distribute the rule to the IXP's own routers as proof that it is operationally viable.

A.1.3 Observation period (30-60 days)

Before applying the policy to members, run it in log-only mode for 30 to 60 days. The objective is:

- Quantify the volume of UDP that would be blocked and from which ASes.
- Identify false positives: legitimate UDP protocols without handshake that members use and that would need to be treated as special cases.
- Build the dataset that will be presented to members. Real numbers, not estimates.

A.2 Phase 2: Build the certification registry

Before speaking to members, the IXP needs the certification mechanism ready. Without it, the proposal is abstract. With it, it is concrete and actionable.

A.2.1 Define certification criteria

The Plane A certification criterion must be simple, verifiable, and non-discriminatory:

- Member implements default-deny inter-AS UDP policy on its border routers toward the IXP.
- Member allows inter-AS UDP only for sessions with verifiable handshake (QUIC Connection ID, WireGuard session, DTLS session ID).
- Member implements rate limiting on DNS/UDP/53 (max 1000 pps as reference).
- Member reports dropped UDP volumes to the IXP monthly (minimum telemetry).

A.2.2 Technical implementation of the registry

The registry can be implemented as a PeeringDB extension or as an IXP-proprietary system:

- Queryable public API: any AS can verify whether a peer is certified before establishing a BGP session.
- Certification process: member declares compliance, IXP verifies with a real traffic sample, certifies and publishes.
- Annual renewal: certification is not permanent, requires revalidation to ensure the member continues to comply.
- Revocation: if a certified member generates anomalous UDP traffic, the IXP can suspend certification with prior notice.

A.2.3 Integration with existing tools

The registry integrates with what already exists:

- PeeringDB: add 'UDP-SAV certified' field to the AS profile, queryable via existing API.
- IXP Manager: certification field in member profile, visible in the portal.
- Route server (while it exists): specific BGP community for certified members, allows automatic differentiated policies.

A.3 Phase 3: The conversation with members

This is the most important part and the most underestimated. The technology is the easy part. Convincing 50 or 100 ISPs to change their border configuration is the hard part. The approach must be operator-to-operator, not regulator-to-regulated.

A.3.1 The argument that works with large ISPs

Large ISPs have technical resources and the DDoS problem hurts them directly. The argument is economic and reputational:

- 'Your sessionless UDP traffic is being used to attack your own customers and your peers' customers. That is your reputational problem, not just the victim's.'
- 'The cost of implementing stateful UDP at your border is a fraction of the cost of a serious amplification incident. A 100 Gbps attack against one of your customers costs you in support, capacity, and customer churn.'
- 'Certified peers will prefer to exchange traffic with other certified peers. If you do not certify, you will be left out of the premium peer table.'
- Show the numbers from the observation period: 'In the last 60 days, your AS originated X Gbps of sessionless UDP that was blocked at our border. This is the traffic you would be eliminating.'

A.3.2 The argument that works with small ISPs

Small ISPs do not have a dedicated security team and the change seems costly. The argument is self-protection and ease:

- 'You are the most frequent victim of amplification attacks. You do not have the bandwidth to absorb them or the budget for scrubbing. This policy protects you, not just others.'
- 'You do not need new hardware. Most equipment you already have supports this. We will help you with the configuration.'

- 'Start in Plane B. No pressure. When you are ready, move to Plane A. Nobody disconnects you.'
- 'Small ISPs that certify first will have an advantage: their peers will know their traffic is clean, which improves peering relationships.'

A.3.3 The argument that works at LACNOG / NANOG

In the operator community context, the approach is technical and regional leadership:

- Present the observation period dataset from the IXP. Real numbers of blocked UDP traffic, distribution by protocol, top origin ASes.
- Show that the IXP2 model already has precedent in RPKI: it is not a new idea, it is the same logic applied to a different layer.
- Propose that PIT Chile be the first IXP in LATAM to implement it and publish the results. That has value for the community and for the IXP.
- Present an Internet Draft at IETF TSVWG as the next formal step. The position paper is the foundation of the draft.

A.4 Phase 4: Escalation and consolidation

A.4.1 Success metrics

The plan needs clear metrics to evaluate progress:

Metric	Baseline	6-month target	12-month target
Plane A certified members	0%	20%	50%
Sessionless UDP volume on fabric	100% (baseline)	-40%	-70%
Members with uRPF enabled	Initial audit	+20pp	+40pp
Members with active Flowspec	Initial audit	+30pp	+60pp
UDP amplification incidents	NOC baseline	-30%	-60%

A.4.2 Program governance

- Technical committee: IXP representatives + 3-5 volunteer members to define updates to certification criteria.
- Annual criteria update: as new UDP protocols with handshake emerge (or new amplification vectors appear), criteria are updated.
- Results publication: public semi-annual report with anonymized statistics on blocked UDP traffic, certification adoption, and incidents avoided.
- MANRS coordination: propose the program as a MANRS IXP framework extension, giving it global reach and institutional credibility.

A.4.3 The step to IETF

Once the IXP has 6-12 months of real implementation data, the next step is to formalize the proposal at IETF:

- Present at IETF TSVWG (Transport Area Working Group) an Internet Draft based on this paper plus real implementation data.
- The real dataset is the most powerful argument: it is not theory, it is evidence that it works and how much traffic it eliminates.
- Coordinate with NIST/Akamai (BAR-SAV authors) to explore whether the verifiable UDP session proposal can complement existing SAV work.
- Goal: in 24-36 months, a RFC or BCP exists defining inter-AS UDP routability criteria, with PIT Chile as the early implementation reference case.

Note on this appendix

This plan assumes that the IXP executing it has the political will to lead it. The technical part is the easiest. The hard part is convincing members, and that is not done with RFCs but with real data, operator-to-operator conversations, and the willingness to be the first to do something uncomfortable. Someone has to be first. This document is the argument for why it is worth being that someone.